

OPERATIONS & CODEBASE INDEX

ShadowTalk AI Enterprise SaaS & File Dictionary

Complete monetization architecture, Stripe integration, white-label tenanting, exhaustive directory catalog, and deployment runbook.

PAYMENT ENGINE

Stripe Subscriptions + Portals

DISTRIBUTION CHANNELS

PWA + Capacitor iOS/Android + Electron

ENTERPRISE MULTI-TENANCY

White-Label Domains & Assets

CODEBASE SCOPE

**119 Core Modules • 123 Chat
Components**

1 SaaS Monetization & Token Economics

ShadowTalk incorporates an enterprise-ready monetization infrastructure spanning monthly recurring subscriptions, micro-transaction token billing, and add-on services (`src/lib/pricingCatalog.ts` and `src/lib/stripe.ts`):

Plan Tier	Pricing	Daily Allowances	Feature Entitlements
Free Tier	\$0 / month	50 messages / day, 3 deep research, 5 images	Standard Groq access, local WebGPU inference, community support.
Pro Tier	\$10 / month	Unlimited messages, 20 deep research, 20 images	Turbo Engine auto-routing, full code executor, priority queue.
Premium Tier	\$15 / month	Unlimited messages, 50 deep research, 50 images	All agents, Cognitive Loop, multi-model consensus, voice copilot.
Elite Tier	\$20 / month	Unlimited everything (fair-use capped)	White-label branding, stealth vault, dedicated tenant SLA, phone support.

Shadow Credits Micro-Billing (useShadowCredits.ts)

To support pay-per-use workflows without forced recurring subscriptions, users can purchase credit packs. High-compute operations dynamically deduct credits: Image Generation (2 credits), Deep Research Synthesis (5 credits), Voice Copilot Session (1 credit/minute).

2 Enterprise Tenancy & Viral Growth

The workspace includes enterprise customization and community virality hooks:

- **White-Label Branding (WhiteLabelBranding.tsx):** Allows enterprise customers to override application logos, navigation titles, primary theme colors, and custom subdomains.
- **Viral Referral Engine (ReferralProgram.tsx):** Captures query parameters via `useReferralTracking.ts`, tracks conversion funnels, and automatically grants free tier upgrades upon successful friend activations.
- **Ethical Behavioral Intelligence (useProactiveAI.ts):** Evaluates visitor hesitation, dwell time, and interaction frequency to offer contextual guidance while adhering to strict privacy guardrails (`PROACTIVE_ETHICS`).

3 Exhaustive Codebase Directory Dictionary

A structured index of the primary repository directories and architectural responsibilities:

Directory Path	Key Sub-Modules	Functional Domain
src/components/chat/	PersonalIDE.tsx , ShadowBrowser.tsx , CognitiveLoopPanel.tsx	Primary interactive chat canvas, modal panels, code canvases, and tool inspectors.
src/lib/turbo/	turboEngine.ts , modelRouter.ts , turboProviders.ts	Intelligent complexity routing algorithm and multi-provider streaming client.
src/lib/memory/	adaptiveMemory.ts , promptInjector.ts	IndexedDB vectorized turn-scoring and contextual prompt memory injection.
src/lib/cyber/	shadowspectre/ (client, prompts, router, scope)	Autonomous security evaluation, vulnerability scanning, and threat intelligence.
src/integrations/firebase/	adapter.ts , firestore.ts , auth.ts , storage.ts	PostgREST query adapter, authentication wrappers, and Firestore synchronization.
functions/src/	index.ts (shared pool, audio mock, drive)	Firebase Cloud Functions v2 handling server-side secrets and rate-limited fallbacks.

4 Build, Testing & Deployment Runbook

Standard operating procedures for compiling, testing, and distributing ShadowTalk AI:

Verification Commands

```
# 1. Static Type Checking (Clean: 0 errors)
npx tsc --noEmit

# 2. Automated Test Suite (29/31 suites passing)
npm test

# 3. Production Vite Compilation
npm run build
```

Deployment Environments

- **Firebase Web Hosting:** Deployed via `firebase deploy --only hosting`. Configured with strict security headers (nosniff, strict-origin-when-cross-origin).
- **PWA Service Worker:** Generated via `vite-plugin-pwa` with Workbox precaching 218 static assets for instant offline launches.
- **Mobile Packaging (Capacitor):** `capacitor.config.ts` defines native application IDs (`com.shadowtalk.ai`), splash screen durations, dark status bar tokens, and keyboard resize behaviors for iOS and Android builds.

Audited Project Health Summary

The project demonstrates exceptional engineering discipline: 100% clean TypeScript compilation with zero type errors, robust code splitting across vendor libraries, complete cryptographic isolation for sensitive data, and seamless fallback to on-device sovereign WebGPU execution.